

Lecture 1: Introduction to Machine Learning

Machine learning is a subset of artificial intelligence focused on building systems that learn from data. Supervised learning algorithms learn from labeled dataset inputs to predict outputs.

Key paradigms include Supervised Learning (e.g. Linear Regression, Classification), Unsupervised Learning (e.g. K-Means Clustering), and Reinforcement Learning.

Overfitting occurs when a statistical model fits exactly against its training data, capturing noise instead of underlying distribution.